



Prepared with Synoptic

Final investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine

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Event overview

The Stage 4 open pit at Bindarra Gold Mine is served by a main ramp. At RL 312 the ramp ran approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, with a dry, well maintained surface. Traffic keeps left, so a descending truck runs on the open, pit side. The RL 312 dewatering spur joins on that side, on the inside of a left-hand bend, at an included angle of 71 degrees. The junction lies 34 m below a convex grade break that crews call the crest.

The spur is a 240 m access road to dewatering bore DW-14, cut under works order WO-26-4471, with earthworks completed on 31/07/2026. It opened to light vehicles on 02/08/2026, when the bore was energised. That day the "ROAD CLOSED - CONSTRUCTION" sign at its mouth was removed, and nothing replaced it. At the time of the occurrence the intersection had no warning sign on either ramp approach, no give-way or stop control, no delineation and no posted speed for the spur. It was absent from the traffic management plan's road layout drawing and was not among the nine intersections where the plan requires a positive radio call. The fleet management system (FMS) in-cab map for Stage 4, last published on 24/07/2026, did not show it.

At about 22:10 on 04/08/2026, a dewatering technician came out of the spur in front of a descending haul truck, reversed back and let it pass. At 23:48 he submitted hazard report HAZ-2026-0917, rated Medium. He wrote that "you cannot see up the ramp from that road" and that "there is no give way on our side", adding: "The trucks are doing forty coming down empty. I do not think they know the road is there." The Dewatering Superintendent closed the report on 06/08/2026. He recorded that the crew had been reminded at the 05/08 pre-start "to take care entering the main ramp and to give way to haul trucks at all times." The rating was reduced to Low on closure, and the report was not referred outside the dewatering department.

The light vehicle driver was a grade control technician employed by Hallorn Survey & Geotechnical and new to Bindarra. He completed Hallorn's two hour online general mining induction on 24/07/2026. On 09/08/2026 the gatehouse issued him a site access card under Hallorn's recognition agreement, after a 20 minute briefing on muster points, speed limits and personal protective equipment. Contractor Management Standard ORR-STD-009 states that the Bindarra light vehicle pit driving assessment "shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted." There is no evidence he completed that assessment or the site-specific supplement. His first night shift began that evening.

The haul truck operator had seven years on 220 t class rigid haul trucks and 14 months at Bindarra. He was on night 10 of a 14 night swing, 8 hours 47 minutes into a 12 hour 30 minute shift, and had slept 6 hours 30 minutes in the preceding day.

At 02:22:41 on 12/08/2026, dispatch called Stage 4 crews back from crib: "All Stage 4 units, crib's done, back to work please." The operator of HT-407 completed one cycle to the ROM pad and started back down empty. At 02:46:21 he called "407, three-sixty crossover, empty, going down through," and dispatch acknowledged. The FMS download shows 39 km/h at RL 384 and 38 km/h at RL 351, descending steadily on the retarder.

The light vehicle driver, in LV-118, was finishing a set of levels at the DW-14 collar for the dewatering department's as-built, his third night at the bore. He said he had driven the spur six or seven times and had never met a haul truck at the intersection or on the ramp below the crest. From the spur's length, his departure from the collar is estimated at about 02:45. His own recollection of about ten to three is later than the recorded collision. The night shift supervisor did not know a light vehicle was in the pit, and no permit, call-in or board existed by which anyone on shift could have known.

At 02:47:06 HT-407 was at RL 317 at 37 km/h, 53 m above the spur mouth. The survey places the first view of the top of a light vehicle's 4.5 m beacon whip, over the crest and above the windrow, at that distance from the operator's eye height, with the body not yet visible. The operator said: "there's a glow on the road, and then the whip comes up over it, and it's a light vehicle, coming out of the side onto my road." He estimated the distance at "forty metres". At 38.5 m above the mouth, 1.4 seconds later, service brake pressure rose from 0 to 620 kPa in 0.4 seconds, the retarder went to maximum and 14 degrees of right steering was applied. Asked about going left, he said: "Right is the wall. Right was the only answer."

The light vehicle driver said he slowed at the mouth, looked right up the ramp, saw nothing and turned right to return to the surface. The turn took the vehicle across the descending lane, right side towards the truck. He estimated his speed at about 25 km/h and, correcting his first account, said he did not believe he came to a complete stop. Neither driver made a radio call before entering the intersection.

At 02:47:09 the collision awareness system on HT-407 logged LV-118 at 21.4 m, closing at 32 km/h. The intersection lay inside the S4-LOAD suppression geofence, so nothing was presented in the cab.

At 02:47:12 the truck's front left struck the right side of LV-118 at 24 km/h, 6.2 m into the ramp from the spur mouth. The light vehicle was pushed 18.4 m down the ramp, going over at 11.2 m where the tyre scuff became a body panel gouge. It came to rest on its left side against the inner windrow, facing down-ramp. The truck stopped 19 m below the intersection at about 02:47:19. The operator said some dust had hung on the ramp but he could not tell whether it made a difference. The light vehicle driver said the air in the spur was clear.

At 02:47:41 the operator called "I've hit a light vehicle on the ramp at three-twelve." At 02:48:01 dispatch stopped all Stage 4 units and held the operator in the cab, releasing him at 02:48:44 with "ramp's clear above and below you." The emergency response team was paged at 02:48. At 02:49:30 the operator reported "He's conscious. He's talking to me. He can't get out, the door's against the ground." The supervisor told him not to move the driver.

The shift's one rostered paramedic was at the surface medical facility. The first response vehicle entered the pit at 02:51 and arrived at 02:54, with the primary survey and extrication plan complete by 02:58. The second responder, required before any extrication, came with the hydraulic rescue tool from the accommodation village 11 km from the gatehouse, arriving at 03:07. The driver was extricated through the rear at 03:14, 27 minutes after the collision, and reached the surface medical facility at 03:41. The site's post-incident review found extrication was delayed by approximately 9 minutes awaiting the second responder and tool. It found this did not affect the clinical outcome, as the injuries were stable and the driver conscious throughout.

The driver had a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was flown to the metropolitan trauma centre at 07:40. LV-118 was destroyed. Its roll-over protection structure was deformed inward at the right rear mounting but not breached, and his fastened seatbelt showed loading marks. HT-407 was damaged at the front left bumper, left mudguard and lower two access ladder steps, and returned to service on 14/08/2026. The operator and shift supervisor returned negative drug and alcohol tests. The Stage 4 pit stood down for 9 hours 15 minutes and the ramp below RL 340 was closed for 22 hours. At 05:10 the Mining Manager directed a positive call on the pit channel before entering any Stage 4 intersection, listed in the plan or not. The occurrence was notified to the State Mines Inspectorate as a reportable injury.

Analysis

The collision followed from one condition. Two vehicles reached the RL 312 intersection together, and neither driver knew the other was there until six seconds remained. This analysis traces how that condition arose, and why it produced a side impact and a roll-over rather than a near miss. The factors are grouped by where they sat, moving from the individuals involved to the organisation above them.

People

Neither driver aware of the other

HT-407 descended the ramp empty at 37 km/h, and nothing told the operator that an intersection lay below the crest. The FMS in-cab map showed none. There was no warning sign on the approach, and no call came over the channel. The collision awareness system raised no alert, although it later logged the light vehicle at 21.4 m. LV-118, for its part, approached a mouth with 38 m of sight line and no give-way control. Its driver had made no call and had no knowledge of the truck. The first sighting open to the truck operator was the beacon over the crest at 53 m, six seconds before the impact. The light vehicle driver had looked up the ramp before the truck was within his view.

None of the evidence shows that either driver knew of the other before that moment. That holds for the pit channel recording, the FMS download, the collision awareness system log and the three interviews. The night shift supervisor did not know a light vehicle was in the pit at all. No permit, call-in or board existed by which he could have known.

The finding is held at extremely likely rather than virtually certain, because the light vehicle's movements before it reached the spur mouth rest on the driver's account alone. Its influence is also extremely likely. The collision followed directly from the convergence, and nothing could have prevented it in the six seconds left.

It is extremely likely that neither driver was aware of the other until the truck came over the crest, and extremely likely that this contributed to the collision.

The positive call that was not made

The pit channel was logged continuously. The investigator listened to the whole period, from the return from crib to two minutes after the collision. No transmission by either vehicle referred to the RL 312 spur, the dewatering bore or a light vehicle anywhere in the Stage 4 pit. HT-407 called at each listed intersection it passed, and each call was acknowledged. It called at the RL 360 crossover going up at 02:28, at the ROM pad turn-off at 02:37, and at the RL 360 crossover coming down at 02:46:21, 51 seconds before the collision. LV-118 made no transmission that shift beyond logging on at 18:04.

No call was required. Clause 4.3.2 of BIN-TMP-004 reads: "At intersections not listed at Appendix B, vehicles shall give way in accordance with clause 4.1. No call is required." The RL 312 spur was not listed. The truck operator, the light vehicle driver and the night shift supervisor each independently described the same practice. Calls are made at the nine intersections on the crib room board, and nowhere else. The supervisor said he had never corrected anyone for not calling elsewhere, and that requiring calls everywhere would make the channel unusable.

A call from either vehicle would have been heard by the other driver, by dispatch and by the supervisor. The dispatcher was already tracking the truck closely, close enough to tell Excavator 2 at 02:46:18 that 407 was "about four minutes off you." Even so, a call gives knowledge and not separation. The two vehicles would still have had to settle who went first across a 41 m sight line, and that is why the influence is put at likely and not higher.

It is virtually certain that neither driver made a positive call before entering the intersection, and likely that this contributed to neither driver being aware of the other.

The light vehicle's entry onto the ramp

On read-back, the light vehicle driver corrected his own account from stopped to slowed, against his own interest. He said he slowed at the mouth, looked right up the ramp, saw nothing and drove out, turning right to go back up the ramp. He put his speed out of the spur at about 25 km/h, and he did not believe he came to a complete stop.

The collision awareness system log places LV-118 at the spur mouth at 02:47:09.2, with the truck 21.4 m above it. The impact came 2.8 seconds later, 6.2 m into the ramp from the spur mouth. The survey gives 38 m of sight line from the mouth to a descending truck, at the driver's eye height. Over the last 25 m of the spur's approach, the 1.6 m inner windrow hides the truck's lower body. On the recorded times, his look came a second or two before the truck came over the crest. Yet the truck was within his 38 m from about 02:47:07.4, nearly two seconds before he reached the mouth.

The rule he described applying was the plan's own. Clause 4.1 reads: "Light vehicles shall give way to heavy vehicles at all times and in all circumstances." Nothing at the mouth told him to do more. There was no stop or give-way control, no stop line, no warning sign and no posted speed. He had not completed the Bindarra light vehicle pit driving assessment, and he had never seen the traffic management plan. He was the injured party, and his entry is analysed as an action taken in those conditions, not as a failing.

Had he stopped at the mouth and looked again, he would very likely have seen the truck and let it pass. The dewatering technician did exactly that on 04/08/2026, by reversing back into the spur. The influence is put at extremely likely rather than virtually certain, because a stop is only as good as the look from it, and the look from this mouth was 38 m.

It is virtually certain that the light vehicle drove out of the spur onto the ramp without stopping, and extremely likely that this contributed to neither driver being aware of the other.

The truck operator's response

The FMS download records the truck's last steady data point at 02:47:06.0, at 37 km/h and 53 m above the spur mouth. That is the point at which the survey puts the first sight of the beacon. From 02:47:07.4, 38.5 m above the mouth, service brake pressure rose from 0 to 620 kPa in 0.4 seconds. In the same movement the retarder went to maximum, and 14 degrees of right steering was applied. The response took 1.4 seconds. That is inside the 1.5 seconds assumed in the stopping distance calculation on which the plan's sight distance rule rests.

From the brake application to the collision, mean deceleration was 0.78 m/s² net of the grade, against the unit's tested 0.80. The impact speed was 24 km/h. With no braking it would have been 37 km/h, so the braking removed about 58 per cent of the impact energy.

The steering mattered too. The braked scuff deviates 1.4 m to the right over its first 20 m, towards the wall and away from the pit edge, where a drop of more than 100 m lay to the operator's left. As a result, the light vehicle came to rest against the pit side windrow rather than being driven through it and over the edge.

The truck stopped 19 m below the intersection, and the operator then called the collision. As dispatch instructed, he stayed in the cab until the ramp was stopped above and below him. He then went to the light vehicle and stayed with the driver until the response team arrived. The truck could not stop whatever he did, but what he did reduced the severity of the outcome.

It is virtually certain that the truck operator braked and steered right within 1.4 seconds of the beacon coming into view, and very likely that this reduced the severity of the collision.

The operator's state on the night

Whether the truck operator was fatigued on the approach was examined and could not be settled. It is assessed as about as likely as not.

Several things pointed towards fatigue. He was on night 10 of a 14 night swing, 8 hours 47 minutes into a 12 hour 30 minute shift. The time was 02:47, inside the window of lowest circadian alertness. He kept a sleep record voluntarily and provided it, and it gives between 5 and 7 hours on each of the ten days before, with 6 hours 30 minutes on the last of them. He described himself as tired "the way you are on night ten". The task on the approach was detection in an unlit pit, and detection is what fatigue degrades first.

Other evidence pointed against impairment. He described the tiredness as ordinary: "Not falling asleep tired ... I was awake and I was driving properly." Every limit in the contractor's fatigue management standard was met. His response of 1.4 seconds was inside the plan's own allowance, and the braking and steering were correct and immediate.

Feeling tired on a tenth night is not the same as being fatigued to a degree that degrades performance. No other evidence reaches the question, and settling it would need a detailed fatigue assessment. None was undertaken, because the outcome could not have turned on it. Braking at 0.78 m/s² from 53 m with a response time of zero still ends in a collision, at about 17 km/h instead of 24 km/h. Whether the operator was fatigued on the approach remains unresolved and is not a finding. The risk built into the roster is analysed under Organisation.

Environment

The sight line at the spur

The survey of 13/08/2026 placed the spur mouth 34 m below a convex grade break where the descent steepens, which crews call the crest. For an operator descending at an eye height of 5.80 m, the survey found two things:

- the top of a 4.5 m beacon whip on a light vehicle approaching the mouth comes into view at 53 m, though "the vehicle's body is not" [Survey SUR-26-1188]
- the spur mouth itself comes into view at 41 m.

In the other direction, a driver at the spur mouth, at an eye height of 1.45 m, first sees a descending truck at 38 m. The survey concluded: "The available sight distance at this intersection is 41 m. The intersection does not meet clause 6.2 on either limb."

Against that, a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade needs 96 m to stop. That figure is calculated with the manufacturer's data and a 1.5 second response. From 37 km/h, braking at the recorded 0.78 m/s² alone takes about 68 m. The plan's 120 m minimum was not met either. The decisive point, though, is that a truck at the posted speed could not stop for anything at this intersection, however alert its operator.

The pit was also dark. There is no fixed lighting anywhere in Stage 4. The light plants at the loading face were 1.1 km away and out of sight behind the crest, and moonset had been at 23:18 the previous evening.

None of this was new. A dewatering technician had reported the same geometry seven days earlier, after coming out of the spur from the same position: "The problem is you cannot see up the ramp from that road. There is a rise in the ramp and a windrow along the side and you are out on the road before you can see anything."

With twice the sight line, the same response and braking would have stopped the truck short of the spur. The influence is still put at very likely rather than virtually certain. A short sight line does harm only once something is in it, and it was the other factors that put something there.

It is virtually certain that the sight line at the spur was 41 m against a 96 m stopping distance, and very likely that this contributed to neither driver being aware of the other.

Airborne dust on the approach

Whether airborne dust delayed the truck operator's sighting of the beacon was examined and could not be settled. The dust itself is likely to have been present. Whether it delayed the sighting is about as likely as not.

Evidence that dust was present. The water cart log records the main ramp as last watered at 01:10, 1 hour 37 minutes before the occurrence. The standing night shift instruction is a pass every 90 minutes, so a pass was 7 minutes overdue. The fixed dust monitor on the RL 328 bench sits 180 m up-ramp of the intersection, above the grade break. It logged PM10 at 142 micrograms per cubic metre in the 02:40 to 02:50 interval, against a shift mean of 88. The truck operator described dust hanging on the ramp. He said the cart had been through earlier and that the road dries out. He was descending behind his own lights into an unlit pit, which is when airborne dust is worst for a driver.

Evidence that limits that conclusion. The reading of 142 was well below the site trigger of 250. It also measures particulate at the monitor head, not what could be seen from either cab. The dust records state: "Nothing at Bindarra measures visibility, and no measurement exists of what could be seen from either cab on the night." The light vehicle driver was firm that the air was clear. He said he could see the ramp surface and the windrow, and that the dust at Bindarra was nothing like other sites he had worked at. His account, though, describes the spur below the windrow, not the ramp above the crest where the truck was. That is why the dust is assessed as likely rather than more certain.

The argument that dust delayed the sighting. Dust hanging in a truck's lights over a crest is the condition in which a driver picks out a beacon late. The operator described dust on the descent.

The argument against. The beacon came into view at 53 m. That is the first point at which the survey says it could be seen at all, even in perfectly clear air, and a sighting at the geometric limit leaves dust nothing to explain. The operator himself declined to say the dust made a difference. A 1.4 second response from that point also leaves no room for a delay to hide in.

Why it could not be settled. The moment of sighting is not directly recorded. It is inferred from the last steady data point and the survey. The evidence cannot exclude dust costing a fraction of a second at the edge of the truck's lights. A measurement of visibility from the cab on the night would settle the question, and none exists. The question remains unresolved and is not a finding.

Equipment

Collision alerts suppressed at the intersection

The collision awareness system on the Stage 4 haul fleet operates at Level 8, operator advisory. Level 9, automated intervention, is available on the fleet but is not enabled at Bindarra. Advisory alerts are raised when a tag enters the 60 m ring at a closing rate above 10 km/h. The exception is inside a suppression geofence. The configuration record states: "Inside a suppression geofence no alert is raised at any range, and the system logs a proximity event only once the range falls below 25 m."

The S4-LOAD geofence was drawn on 14/03/2025 around the Stage 4 loading face. Its purpose was to stop nuisance alerts between the excavator, the trucks under it and nearby service vehicles, where the alert rate had made the system unusable. That purpose was sound. On 09/06/2026, when the loading face advanced, the contractor's technical services group extended the geofence northward as a system configuration task. For the first time, this carried its boundary up the main ramp, to a point 180 m up-ramp of where the RL 312 spur was later cut. The system configuration gives the new boundary but not the old one, so the size of the extension is not known. The extension was not recorded in the traffic management plan, nor reviewed against it. About seven weeks later the spur was cut, and it joined the ramp inside the suppressed area.

At 02:47:09.2 the system logged LV-118 at 21.4 m, closing at 32 km/h, with the alert suppressed. Nothing was presented in the cab.

The fleet statistics for the twelve months before the occurrence show a pattern that nobody could see:

- HT-407 raised 1,684 advisory alerts, every one outside the geofence and none inside it.
- Across the fleet, 68 proximity events inside the geofence were suppressed.
- Nine of those 68 involved a light vehicle tag on the main ramp below the boundary, and all nine are dated after 09/06/2026.

The suppressed events were stored, and anyone who queried the system could have seen them. The technical services group's standing report, however, covers alert counts and system health and does not report suppressed events. No training on site covers the suppression geofences. The truck operator and the night shift supervisor each said, unprompted, that they did not know the system was off on the ramp.

The existence of this factor is put at extremely likely rather than virtually certain. The configuration and logs were downloaded sealed from the system itself. The operators' lack of knowledge, by contrast, rests on the two accounts and on the absence of any training or notice.

The alert itself would have bought nothing. Unsuppressed, it would have sounded at about 02:47:06.5, half a second after the beacon came into view and about a second before the brakes went on. It would only have told the operator what he was already reacting to, and the system carries no intervention.

The influence that remains is the operator's expectation. Before the June extension, the system was live on the ramp below the crest. It raised no alert there at night because nothing was there. From June it was silent whatever was on the road, and nobody told the operators. The truck operator wrote in his own report, before he was asked, that he "took that to mean the pit was empty when it was quiet". He came over the crest with that expectation, and nothing else told him an intersection lay below.

Against this, the expectation is not shown to have changed anything he did. He was descending at the speed every operator uses on that ramp. Afterwards, the remedy he named was a sign, not the system. The influence is accepted as likely, at the standard of proof and no higher, and both readings stand.

It is extremely likely that collision alerts were suppressed at the intersection by a geofence the operators did not know about, and likely that this contributed to neither driver being aware of the other.

The impact and the roll-over

The spur joins the ramp at an included angle of 71 degrees on the descending truck's left, on the open side of the road. LV-118 was turning right to go up the ramp. That took it across the

descending lane with its right side towards the truck, so the geometry made a side impact the likely result of any conflict.

The impact occurred at 02:47:12.0 at 24 km/h. It was 6.2 m into the ramp from the spur mouth, and 3.1 m out from the toe of the inner windrow. The truck's accelerometer logged 1.9 g longitudinal. Damage to LV-118 was concentrated on its right side from the B pillar forward. Damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line. Right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact position.

The light vehicle was pushed 18.4 m down the ramp. At 11.2 m the tyre scuff became a body panel gouge, and that is where the vehicle went over. It came to rest on its left side against the inner windrow, facing down-ramp. Extrication was through the rear.

The occupant sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. These are the injuries a side impact and roll-over produce when the structure holds and the restraint is worn. Both held here. The roll-over protection structure deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened, with loading marks on the webbing. The beacon lamp was tested after recovery and worked, though the whip had sheared at its base in the roll-over.

It is virtually certain that the haul truck's front left struck the right side of the light vehicle at 24 km/h, and that the light vehicle was pushed about 18 m down the ramp and rolled onto its left side. It is virtually certain that both contributed to the driver's injuries.

Procedures

The positive-call list

Clause 4.3.1 of BIN-TMP-004 requires that "A positive call shall be made on the relevant pit channel before entering any intersection listed at Appendix B, and entry shall not be made until the call is acknowledged or the channel is confirmed clear."

The control is sound. The way an intersection reaches the list, however, is circular. The note to Appendix B provides that the appendix "is reviewed and reissued at the annual review of this plan", and that "An intersection is added to this appendix where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication." Schedule 1 is itself reissued only at the annual review. No provision allows an intersection to be added to either document between reviews. Nor does any provision require a new road to be referred to the plan's owner when it is cut.

The RL 312 spur was on neither document. The Appendix B board in the Stage 4 crib room was the 12/11/2025 issue. The next date on which the spur could have been listed was 12/11/2026, three months after the collision.

The gap is not confined to this spur. Annual reviews were completed on 14/11/2023, 10/11/2024 and 12/11/2025. In the plan's eleven year history, no out-of-cycle review has been held under clause 9.1. Fourteen new roads have been cut in the Stage 4 pit since the last review, and eleven of them were never referred to the plan's owner. A search of the hazard reporting system for the 24 months to 11/08/2026 returned four reports, including HAZ-2026-0917. The other three are from 03/2025, 09/2025 and 02/2026. Each describes a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B, at three different locations. All four were closed within three days, with an entry recording that the crew had been reminded to take care and that heavy vehicles have right of way. None was referred to the plan's owner. None led to a change in Appendix B, Schedule 1 or any physical control.

Had RL 312 been listed, the call would very likely have been made, and the channel was open with the light vehicle on it. The influence is put at very likely rather than virtually certain, because a call made and not acknowledged still leaves a driver deciding whether to go.

At 05:10 that morning, the Mining Manager directed a positive call before entering any intersection in the Stage 4 pit, listed or not, until further notice.

It is virtually certain that the plan required positive calls only at listed intersections, with intersections added only at the annual review, and very likely that this contributed to neither driver making a positive call.

The guideline under which the list is made

The plan is written within Mines Safety Guideline MSG-14, made under regulation 6.22 of the Mining Operations Safety Regulations 2018. Clause 3.2 requires the plan to be "reviewed at intervals not exceeding twelve months, and following any significant change to the mine's road network, vehicle fleet or operating pattern." The guideline does not define "significant change", nor does it give an example. Clause 5.1 requires the operator to "nominate the intersections, crossings and other interaction zones at the mine at which positive communication between vehicles is required before entry". It sets no minimum and no basis for the nomination. Clause 3.5 provides that the plan "is a site document. It is not approved by, and need not be submitted to, the Inspectorate, except on request following an occurrence or an inspection."

On its face, Bindarra's plan met the guideline. It had been reviewed within twelve months and it nominated nine interaction zones, so the mine was not in breach. Clause 5.4, which offers five alternative means of making a light vehicle detectable, is context and not part of this factor.

The point concerns the shape of the discretion the guideline leaves, not any failure by the Inspectorate. Both open terms are left to a site document that is reviewed once a year and that no one outside the mine looks at. Had the guideline defined a significant change, adding a road to an operating haul route would have required the plan to be reopened. Had it set a basis for nomination, an intersection that failed the sight distance rule would have had a claim on the list. Either way, the eleven Stage 4 roads never referred would then have had a defined route onto it between reviews.

The argument against is real. The guideline stopped nobody. The mine could have held an out-of-cycle review under its own clause 9.1 at any time, and MSG-14 does not prevent a mine from listing every intersection. Read that way, nothing inside the mine ever put the question to the plan's owner, and the guideline is background.

The influence was accepted as likely, on the balance and at the standard of proof and no higher. That says only that the guideline made inaction the default, and it is not presented as settled. The guideline also bears on the listing path only. Whether a works order is exempt from management of change is set by the mine's own procedure.

It is virtually certain that the guideline set no review trigger and no basis for the positive-call list, and likely that this contributed to positive calls being required only at listed intersections added at the annual review.

Intersection controls at RL 312

Two clauses of BIN-TMP-004 govern intersection controls, and they reach different sets of intersections. Clause 6.2 applies "At every intersection", listed or not. It requires available sight distance of not less than the stopping distance of the largest vehicle at the posted speed, and in no case less than 120 m. Its second limb reads: "Where this cannot be achieved, the intersection shall be controlled by a stop control, signalised control or traffic controller." With 41 m against 96

m, that limb applied at RL 312. Clause 6.4 requires an intersection warning sign 50 m in advance on each approach, a give-way control on the minor road and delineation at the mouth. It applies, however, only at intersections "shown on the road layout drawing at Schedule 1", and the spur was not on that drawing.

At the time of the occurrence, the intersection carried:

- no intersection warning sign on either ramp approach
- no give-way or stop control at the spur mouth
- no delineation, reflective marker posts or windrow gap markers
- no posted speed for the spur itself.

The "ROAD CLOSED - CONSTRUCTION" sign was removed on 02/08/2026, the day the spur was first used, and nothing was installed in its place. Seven days before the collision, hazard report HAZ-2026-0917 named both absences: "There is no sign on the ramp telling the trucks the road is there and there is no give way on our side."

The argument that the missing controls mattered. A stop control at the spur mouth would have held the light vehicle while the truck came into view. A warning sign on the truck's approach would have told the operator that an intersection lay below the crest. The operator said: "If there'd been a sign on my side of that crest I'd have been at twenty and I'd have been looking for him."

The argument against. From the mouth, the windrow and grade break give 38 m. That is about four seconds of a truck at 37 km/h, a thin margin even from a standstill. The driver was already slowing and looking under a give-way rule. Whether a sign would have slowed the truck rests on the operator's own view after the event. A sign at an intersection absent from the in-cab map would also have been a single cue.

A full stop at the mouth would very likely have shown the driver the truck. The influence is nonetheless put at likely, below the other findings. The margin from a stop at 38 m is thin, and the effect of a sign on the truck rests on the operator's own account.

It is virtually certain that there was no stop control or warning sign at an intersection that failed the plan's sight distance rule, and likely that this contributed to the light vehicle driving out of the spur onto the ramp without stopping.

Organisation

The spur opened under a works order

Management of change procedure ORR-PRO-022 clause 5.2 requires a management of change record "before any of the following: a change to the mine's road network". Clause 5.3 then excludes "routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced." Under clause 5.4, a management of change record requires the change owner to make three referrals. The change goes to the traffic management plan owner, to the survey department for the fleet management system road network, and to the contractor's technical services group for the collision awareness system configuration. The works order form at Schedule 4 has no equivalent fields and no referral of any kind.

Works order WO-26-4471 was raised by the Mining Engineer on 28/07/2026 and approved by the Mining Superintendent on 29/07/2026. It cut 240 m of access to bore DW-14 at RL 312. Its classification reads: "Routine earthworks within the approved Stage 4 design. No new plant or vehicle type. Management of change not required, ORR-PRO-022 clause 5.3." Its handover field reads "N/A". It carried no scope for signage, survey pickup of the new intersection, referral to the plan, republishing of the road network or review of the geofences.

None of the three referrals was made, and the absence of each shows in the place it would have reached:

- The FMS in-cab map, last published on 24/07/2026, was still current on the night and showed no intersection at RL 312.
- The plan's owner was never told of the intersection.
- The S4-LOAD geofence already covered the junction, and nobody responsible for it was asked to look.

The pattern holds across Stage 4. Of the fourteen new roads cut there since the last plan review, three were raised under a management of change record, and those three were referred to the plan's owner. Eleven were raised under a works order, and none of them was.

Had a management of change record been raised, the referral to the plan's owner would have led to a survey showing the intersection failing clause 6.2. A stop control and the clause 6.4 signs would have followed. The record would also have gone to the technical services group, who configure the geofence the intersection was cut inside. The influence is put at very likely rather than virtually certain, because a referral is not itself a control. The plan's owner might have left the spur to the annual review, and no out-of-cycle review has been held in the plan's eleven years. A referral also could not have changed the listing rule. This factor therefore bears on the intersection controls and the geofence, not on the positive-call list.

It is virtually certain that the spur was opened under a works order and so was referred to no one, and very likely that this contributed to the absence of a stop control and warning signs at the intersection and to the suppression of collision alerts there.

Access to the operating pit

Contractor Management Standard ORR-STD-009 clause 7.3 allows a contractor's own general induction to be accepted as equivalent where the contractor holds a current recognition agreement. Hallorn Survey & Geotechnical has held one since 2023. Clause 7.4 sets the limit: "Equivalence under 7.3 does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted." Clause 7.5 makes the engaging department responsible for verifying 7.4 before a site access card is issued.

The standard says the right thing. What failed is that the only place the verification could happen has no means of doing it. The gatehouse issues cards against the recognition agreement alone, and it holds no entry for either the supplement or the assessment. Nothing in the access card system distinguishes a person who may drive in an operating pit from one who may not.

The driver's card was issued on 09/08/2026 after a 20 minute gatehouse briefing. His training file shows the Hallorn online module of 24/07/2026 and no pit driving assessment. He had never heard of the assessment, and he had never seen BIN-TMP-004. [NOT ESTABLISHED: whether the Bindarra site-specific supplement was issued to the driver.] This is not a failure by the driver.

The argument that the assessment would have mattered. Had the gatehouse checked, the driver would not have been carded to drive in an operating pit until he had done the assessment. The assessment is where a light vehicle driver learns how the pit's traffic rules work in practice. That covers giving way to haul trucks, approaching an intersection he cannot see up, and when calls are made. He said that if he had been told to call, he would have called.

The argument against. He was already applying the rule the assessment teaches first, giving way to heavy vehicles at all times, and he slowed and looked before he went. The plan requires no call at an unlisted intersection, so the assessment would not have taught him to call at RL 312.

What the assessment adds is how to approach a mouth he cannot see out of, which is to stop and look again rather than slow. The influence is put at likely, below the other findings. He was already applying the assessment's first rule, and the assessment's content was not available. [NOT ESTABLISHED: the content of the Bindarra light vehicle pit driving assessment.]

It is virtually certain that site access cards were issued with no check of the pit driving assessment, and likely that this contributed to the light vehicle driving out of the spur onto the ramp without stopping.

The haul roster

The contractor's Stage 4 haul roster is 14 nights on and 7 off, from 18:00 to 06:30. That is a 12 hour 30 minute shift with a 30 minute crib. The contractor's fatigue management standard permits up to 14 consecutive night shifts with a minimum 10 hour break. Every limit was met, and none of the operator's previous nine shifts was extended beyond 12 hours 40 minutes. The night shift supervisor works the same pattern as the crews he supervises.

The argument for an influence is that a shorter roster would not have put the operator at the wheel at 02:47 on a tenth consecutive night, where slower detection and response are expected. The recorded data contradicts that mechanism. The 1.4 second response cost about 7 km/h of impact speed, not the collision. The download also shows no decision or omission that a roster could have worked through. The influence is therefore unlikely, and that conclusion is reached on the evidence and is not a dismissal.

The roster remains important. On every swing, it puts every Stage 4 haul operator at the wheel of a 220 tonne vehicle in the small hours of a tenth, twelfth and fourteenth consecutive night. At an unlit, unlisted intersection, the defence is the operator's own detection and response. At an intersection with a longer sight line than RL 312, a slower response could be the difference between stopping and not.

It is virtually certain that haul operators were rostered for fourteen consecutive 12.5 hour night shifts. It is unlikely that this contributed to the occurrence, and likely that it increased risk.

Night shift extrication

The Bindarra night shift was rostered with one qualified mines rescue paramedic, based at the surface medical facility. A second responder must be present before any extrication begins. That responder is called in from the accommodation village, 11 km from the gatehouse, in the second response vehicle, which is garaged there and carries the hydraulic rescue tool. [NOT ESTABLISHED: whether the same emergency response arrangement applies on the site's other night shifts.]

On the night, the paramedic's primary survey and extrication plan were complete by 02:58. The second responder arrived with the tool at 03:07, and the driver was extricated at 03:14. The post-incident review found the extrication delayed by approximately 9 minutes. It recorded that the delay did not affect the clinical outcome, because the injuries were stable and the patient was conscious throughout. The collision and the injuries were complete before the team was paged, so the influence is unlikely.

The importance is structural rather than clinical. With a fuel fire, an unconscious casualty, or an airway that needed the vehicle opened in the first ten minutes, the same nine minutes could produce a different outcome. The night shift extrication capability is one person plus a drive from the village, and no part of the arrangement scales with what has happened.

It is virtually certain that night shift extrication waited on a second responder called from the village. It is unlikely that this contributed to the occurrence, and very likely that it increased risk.

Hypotheses tested and excluded

Two hypotheses were tested and excluded.

The first was a wet or slippery ramp surface, assessed as very unlikely to have existed. No precipitation was recorded on the night, and the last rainfall at the site was 26 days earlier. The ramp had last been watered at 01:10. When examined at 06:20, it was dry throughout, with no damp patches, ponding or greasy film at or above the intersection. The braked scuff ran continuously for 58 m with no sign of a locked or sliding wheel.

The second was that the service brakes delivered less than their tested deceleration, also assessed as very unlikely to have existed. Brake performance tests on 03/08/2026 and 14/08/2026 gave 0.80 and 0.82 m/s² net, and the 0.78 achieved during the occurrence lies inside their normal scatter. Service brake pressure rose from 0 to 620 kPa in 0.4 seconds, and the retarder was at maximum throughout. No brake, retarder or park brake defect had been raised in the preceding twelve months, and the pre-start check for 11/08/2026 recorded no defects.

The truck could not stop in the distance available, and the sight line explains why, not the surface or the brakes.

Findings summary

A finding stated without a likelihood term was established as fact.

Contributing factors

- It was virtually certain that the RL 312 spur was opened under works order WO-26-4471 as routine earthworks excluded from management of change, so nothing was referred to anyone, and very likely that this contributed to the collision. (Safety Issue)
- Collision alerts at the spur were very likely suppressed by the S4-LOAD geofence, which operators knew nothing of. It was likely that this contributed, held at that level and no higher, through the operator's expectation that a quiet system meant an empty ramp rather than through the alert itself. (Safety Issue)
- At the spur mouth the sight line was 41 m against a 96 m stopping distance for a 220 t class truck at the posted speed, a condition that was virtually certain and that very likely contributed.
- Mines Safety Guideline MSG-14 virtually certainly set no review trigger and no basis for the positive-call list, and it was likely, held at that level and no higher, that this contributed by leaving the listing of a new intersection to an annual site review, with Bindarra's plan meeting the guideline on its face. (Safety Issue)
- Under the traffic management plan, positive calls were virtually certainly required only at listed intersections, added only at the annual review, and this very likely contributed. (Safety Issue)
- It was virtually certain that neither driver made a positive call before entering the intersection, where the plan required none, and likely that this contributed.
- Access cards were virtually certainly issued with no check of the pit driving assessment, and it was likely that this contributed to the light vehicle's entry onto the ramp, held at that level because no copy of the assessment's content was available. (Safety Issue)
- It was virtually certain that the RL 312 intersection had no stop control or warning sign although it failed the plan's sight distance rule, and likely that this contributed. (Safety Issue)
- After slowing and looking up the ramp, the light vehicle driver drove out of the spur onto the ramp without stopping, which was virtually certain, and it was very likely that this contributed.
- Neither driver was very likely aware of the other until the truck came over the crest, and this condition very likely contributed.
- The haul truck's front left struck the right side of the light vehicle at 24 km/h, which was virtually certain, and the impact virtually certainly contributed to the injuries.
- That the light vehicle was pushed 18 m down the ramp and rolled onto its left side was virtually certain, and its contribution to the injuries was virtually certain.

Other factors that increased risk

- Haul operators were virtually certainly rostered for fourteen consecutive 12.5 hour night shifts. Although this was not found to have contributed to the collision, it was likely that the roster increased risk. (Safety Issue)
- It was virtually certain that night shift extrication waited on a second responder called from the accommodation village 11 km from the gatehouse. Although this was not found

to have affected this occurrence or its outcome, it was very likely that the arrangement increased risk. (Safety Issue)

Other findings

- The investigation examined whether the ramp surface was wet or slippery at and above the intersection and ruled it out, because the surface was found dry and the truck achieved its dry-surface deceleration.
- The investigation examined whether the service brakes delivered less than their tested deceleration during the stop and ruled it out, because the recorded 0.78 m/s² net lay within the scatter of tests at 0.80 and 0.82 before and after the occurrence.
- The truck operator braked and steered right within 1.4 seconds of the beacon appearing, which reduced the impact speed from 37 km/h to 24 km/h and kept the light vehicle running down the ramp against the inner windrow rather than towards the open pit edge.

Appendices

Appendix A: Test summary of factors examined

Each factor was tested in three stages:

- Existence: whether the condition or event existed.
- Influence: whether it influenced the occurrence or its outcome.
- Importance: for a factor that did not pass the influence test, whether it was still important to safety.

The likelihood terms in the table are those recorded against each test. The factors are grouped by lane, the level at which each sat.

Factor	Lane	Existence	Influence	Importance	Category
Guideline sets no review trigger and no basis for the positive-call list	External Influences	Pass, virtually certain	Pass, likely	Not tested	Contributing factor (Safety Issue)
Spur opened under a works order, so nothing was referred to anyone	Organisational Influences	Pass, virtually certain	Pass, very likely	Not tested	Contributing factor (Safety Issue)
Access cards issued with no check of the pit driving assessment	Organisational Influences	Pass, virtually certain	Pass, likely	Not tested	Contributing factor (Safety Issue)
Haul operators rostered for fourteen consecutive 12.5 hour night shifts	Organisational Influences	Pass, virtually certain	Fail, unlikely	Pass, likely	Other factor that increased risk (Safety Issue)
Positive calls required only at listed intersections, added only at the annual review	Risk Controls	Pass, virtually certain	Pass, very likely	Not tested	Contributing factor (Safety Issue)
Collision alerts suppressed at the spur by a geofence operators knew nothing of	Risk Controls	Pass, extremely likely	Pass, likely	Not tested	Contributing factor (Safety Issue)
No stop control or warning sign at an intersection failing the sight distance rule	Risk Controls	Pass, virtually certain	Pass, likely	Not tested	Contributing factor (Safety Issue)
Night shift extrication waits on a second responder called from the village	Risk Controls	Pass, virtually certain	Fail, unlikely	Pass, very likely	Other factor that increased risk (Safety Issue)
Sight line of 41 m at the spur against a 96 m stopping distance	Local Conditions	Pass, virtually certain	Pass, very likely	Not tested	Contributing factor

Fictional case study for training. Not a real investigation.

Airborne dust on the ramp above the intersection during the truck's approach	Local Conditions	Pass, likely	Inconclusive, about as likely as not	Not tested	Unresolved, not a finding
Haul truck operator fatigued during the approach to the intersection	Local Conditions	Inconclusive, about as likely as not	Not tested	Not tested	Unresolved, not a finding
Ramp surface wet or slippery at and above the intersection	Local Conditions	Fail, very unlikely	Not tested	Not tested	Excluded
Neither driver made a positive call before entering the intersection	Individual Actions	Pass, virtually certain	Pass, likely	Not tested	Contributing factor
Light vehicle drove out of the spur onto the ramp without stopping	Individual Actions	Pass, virtually certain	Pass, extremely likely	Not tested	Contributing factor
Truck operator braked and steered right within 1.4 seconds of the beacon appearing	Individual Actions	Pass, virtually certain	Pass, very likely (a decrease in risk)	Not tested	Positive event
Neither driver aware of the other until the truck came over the crest	Occurrence Events	Pass, extremely likely	Pass, extremely likely	Not tested	Contributing factor
The haul truck's front left struck the right side of the light vehicle at 24 km/h	Occurrence Events	Pass, virtually certain	Pass, virtually certain	Not tested	Contributing factor
The light vehicle was pushed 18 m down the ramp and rolled onto its left side	Occurrence Events	Pass, virtually certain	Pass, virtually certain	Not tested	Contributing factor
Service brakes delivered less than their tested deceleration during the stop	Occurrence Events	Fail, very unlikely	Not tested	Not tested	Excluded

The influence result for the truck operator's response measures how much it reduced the harm. It does not measure a contribution to the collision.

Two matters were examined and could not be settled. Neither is a finding.

The evidence supported the presence of airborne dust on the ramp above the grade break as likely. The water cart pass was 7 minutes overdue. The dust monitor on the RL 328 bench logged PM10 at 142 micrograms per cubic metre against a shift mean of 88. Against this, the reading was well below the site trigger of 250. It measured particulate at the monitor head, not what either driver could see. Whether the dust delayed the truck operator's first sighting was about as likely as not. The beacon was in view from 53 m, the first point at which the survey says it could be seen at all. The 1.4 second response from that point left no room for a delay. That sighting point is inferred from the last steady data point and the survey, however, and it was not recorded. A loss of a fraction

of a second at the edge of the truck's lights therefore cannot be excluded. Nothing at Bindarra measures visibility, and no recorded sighting exists to settle the matter.

Whether the truck operator was fatigued to a degree that degraded his performance was about as likely as not. His duty and the hour pointed towards fatigue. He was on night 10 of 14, at 02:47, after 6 hours 30 minutes of sleep. His own account pointed the other way, and so did his recorded response. Settling the question would need a detailed fatigue assessment, which was not undertaken. Even if fatigue had been established, it could not have influenced the collision. Braking at 0.78 m/s^2 from 53 m with no response time at all would still have ended in a collision, at about 17 km/h instead of 24 km/h.

Appendix B: Excluded hypotheses

Ramp surface wet or slippery at and above the intersection

This hypothesis was tested because an 18 m push down a 9 per cent grade raises a question about the surface. Several sources ruled it out:

- The met station recorded no precipitation on the night. The last rainfall at the site was 26 days before the occurrence.
- The water cart's last pass on that section was at 01:10, 1 hour 37 minutes before the occurrence. The standing interval is 90 minutes, so the surface had been drying for longer than the interval rather than being freshly watered.
- The investigator examined the ramp at 06:20 and found it dry throughout. There were no damp patches, no ponding and no greasy film at or above the intersection.
- The braked tyre scuff ran continuously for 58 m with no sign of a locked or sliding wheel.
- The download gave a mean deceleration of 0.78 m/s^2 net. The truck's own tests on the same dry grade gave 0.80 and 0.82.

No source gave any evidence of a wet or slippery surface. That the truck could not stop in the distance available is a matter of the sight line at the spur, not of the surface.

Service brakes delivered less than their tested deceleration during the stop

This hypothesis was tested because a haul truck that does not stop in the distance available raises the question of its brakes. The evidence ruled it out:

- The monthly brake performance test on 03/08/2026 gave a mean deceleration of 0.80 m/s^2 net, empty, on a 9 per cent descending grade.
- The same test, repeated on 14/08/2026 before the unit returned to service, gave 0.82.
- The download of the occurrence gave 0.78 over 4.6 seconds of braking, from 37 km/h to 24 km/h. This lies within the normal scatter of the two tests.
- Service brake pressure rose from 0 to 620 kPa in 0.4 seconds, and the retarder was at maximum throughout.
- No brake, retarder or park brake defect had been raised against the unit in the preceding twelve months.
- The pre-start check sheet for 11/08/2026 was complete and recorded no defects.

The braking system performed to specification before, during and after the occurrence.

Appendix C: Sources of evidence

Interviews

Role	Form	Date
Haul truck operator	Recorded and transcribed	Completed by 21/08/2026

Night shift supervisor	Written up and read back to the interviewee	Completed by 21/08/2026
Light vehicle driver	Written up and read back to the interviewee	Completed by 21/08/2026

[NOT ESTABLISHED: the date of each individual interview]

The dewatering technician's account of 04/08/2026 came from his written hazard report, HAZ-2026-0917. He was not interviewed.

Documents, data and examinations

Source	Date or period	Content
Occurrence notification ORR-OCC-2026-0332	Raised 12/08/2026 at 03:26	Place, time, vehicles, injuries, damage and actions taken on the night
Scene examination INV-2026-0332-SCENE	12/08/2026, before recovery	Braking marks, impact position, push and roll-over, restraint and roll-over protection, signage and delineation at the RL 312 intersection
Tarrowen Mining Services technical services report TRW-TS-2026-114	Issued 18/08/2026	FMS download for HT-407: speed, response time, braking achieved, impact speed and the in-cab road network
CAS configuration and event log, HT-407	Extracted 12/08/2026	System level, S4-LOAD geofence configuration and the suppressed proximity event at 21.4 m
Tarrowen technical services fleet CAS event log extract	12 months to 11/08/2026	Advisory alert and suppressed event statistics for HT-407 and the fleet
Pit channel radio transcript RAD-2026-0332	02:20 to 02:52 local (18:20 to 18:52 UTC), transcribed 17/08/2026 and verified by a second investigator	Calls made at listed intersections, the absence of any call about the spur, and emergency response times
Mine survey pickup SUR-26-1188	13/08/2026	Intersection geometry, sight distances from both vehicles and stopping distance
Hazard report HAZ-2026-0917 and closure entry	Submitted 04/08/2026, closed 06/08/2026	The technician's report of the sight line and missing controls at the spur
Works order WO-26-4471	Raised 28/07/2026, completed 31/07/2026	Construction of the RL 312 dewatering access and its classification as routine earthworks
Bindarra Traffic Management Plan BIN-TMP-004 revision 7	Issued 12/11/2025	Clauses 4, 6 and 9 and Appendix B
Document register for BIN-TMP-004 and Orrabinda reporting system search	24 months to 11/08/2026	Plan review history, roads referred to the plan's owner and related hazard reports
Management of Change procedure ORR-PRO-022 revision 3	In force on the day	Clauses 5.2 to 5.4 and Schedule 4
Contractor Management Standard ORR-STD-009 revision 5	In force on the day	Clauses 7.3 to 7.5
Mines Safety Guideline MSG-14 revision 4, made under Mining Operations Safety Regulations 2018 r 6.22	In force on the day	Clauses 3.2, 3.5, 5.1 and 5.4

Tarrowen training file, Hallorn training file and Bindarra gatehouse access card register	As held at the time of the occurrence	Training, competency and induction of both drivers
Tarrowen roster and site access records, and the haul truck operator's sleep record, provided voluntarily	Swing ending with the occurrence	Roster, duty and sleep
Met station, water cart log and fixed dust monitor records	Night of 11 to 12/08/2026	Light, surface, weather and dust
Tarrowen asset register and Orrabinda light vehicle register	Obtained 13/08/2026	Maintenance, inspection and brake test records for both vehicles
Bindarra emergency response log and post-incident review	Obtained 26/08/2026	Emergency response times and the site's review of the extrication

Appendix D: Investigation activity

The investigation log recorded its entries in UTC. They are given below in local time, with the logged UTC time in brackets.

- **12/08/2026 at 04:15 (11/08/2026 20:15 UTC).** The investigator was notified by the Night Shift Superintendent. The occurrence was assessed as a reportable serious injury warranting investigation. The investigation was opened, its scope set and its folder created. Statutory notification was made to the State Mines Inspectorate.
- **12/08/2026 at 12:00 (04:00 UTC).** The scene was attended before recovery. Both vehicles were photographed and measured in position and the marks recorded. The contractor took the fleet management and collision awareness downloads and handed them over sealed. The haul truck operator and the shift supervisor were tested for drugs and alcohol, with negative results. A site-wide direction had been issued overnight requiring a positive call at every Stage 4 intersection, listed or not.
- **13/08/2026 at 15:30 (07:30 UTC).** A survey pickup of the intersection was requested and received. It gave an available sight distance of 41 m against a plan minimum of 120 m and a stopping distance of 96 m. Works order WO-26-4471 was located, with management of change recorded as not required and the handover field reading N/A. Hazard report HAZ-2026-0917 and its closure entry were obtained from the reporting system.
- **21/08/2026 at 14:40 (06:40 UTC).** Interviews with the haul truck operator, the night shift supervisor and the light vehicle driver were completed. Collision awareness statistics were requested following the accounts of the operator and the supervisor.
- **28/08/2026.** A preliminary update was issued.
- **01/09/2026 at 10:00 (02:00 UTC).** The investigation confirmed that Appendix B of the traffic management plan could not be added to between reviews. An intersection reaches the list only through a drawing reissued once a year, and nothing requires a new road to be referred to the plan's owner. The matter was raised with the Mining Manager.

Distribution

Title	Name
General Manager, Bindarra Gold Mine	T. Brandner
Registered Manager (statutory)	S. Kirkhope
Mining Manager and owner of BIN-TMP-004	H. Quaile
Manager Health and Safety, Orrabinda Resources	L. Farquhar

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Project Manager, Tarrowen Mining Services	D. Mokoena
Superintendent Dewatering, Orrabinda Resources	P. Nyaguse